

Post Mining Land Use | Rezoning | Macquarie Coal Complex Pilot Project

Dam Engineering Report

Department of Planning, Housing and Infrastructure

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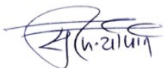
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Executive summary

This Dam Engineering Report has been prepared by Aurecon on behalf of the Department of Planning, Housing and Infrastructure (DPHI) to inform the State-led rezoning of the Macquarie Coal Complex (the Precinct) for post-mining land use. The report provides a risk-informed planning framework to guide future land use, infrastructure design, and staging decisions relating to the existing Tailings Storage Facility (TSF), which remains regulated as declared dam.

The TSF represents a key constraint to short and medium development, due to the residual risk associated with potential dam failure. Dambreak modelling has identified a downstream inundation zone, within which development has the potential to increase risk to people (Population at Risk and Potential Loss of Life). While current risk levels are well below regulatory thresholds, land use planning must not increase these risks, in accordance with NSW dam safety requirements. A staged and adaptive planning approach is therefore required.

Subsequently, post mining land use planning within and downstream of the TSF must adopt a risk-based framework that:

- Avoids increasing population exposure within the dambreak inundation zone
- Requires periodic reassessment of dambreak risk as site conditions change
- Aligns developable land to demonstrated risk reduction
- Ensures compatibility between land use and the evolving residual risk profile of the TSF.

Key post mining land use implications

- Development constraints: Land within the dambreak inundation zone is constrained while the TSF remains a declared dam. High-intensity or sensitive land uses which increase societal risk (e.g. residential and industrial) are not appropriate in the short to medium term.
- Permissible interim uses: Low-intensity, transient or non-sensitive uses (such as open space, conservation or passive recreation) may be considered, subject to maintaining flood conveyance and not increasing risk.
- Infrastructure design: Infrastructure (including roads and utilities) within or crossing the inundation zone must be designed above modelled inundation levels and resilient to flood impacts.
- TSF landform constraints: Development directly on the TSF is limited in the short and medium term due to ongoing settlement and geotechnical constraints. Low impact uses that do not compromise the integrity of the rehabilitated landform and capping are appropriate.

Staging and future development potential

The development potential of the Precinct will increase over time as rehabilitation progresses and dam-related risks reduce. This is expected to occur through:

- Progressive lowering of water levels within the TSF (phreatic surface reduction)
- Completion of capping and landform stabilisation
- Updated dambreak and consequence assessments demonstrating reduced inundation extent and risk

As these milestones are achieved, land currently constrained may be progressively released for more intensive development, subject to updated technical verification.

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Main report



1 Introduction

This Dam Engineering Report (this Report) has been prepared by Aurecon on behalf of the Department Planning, Housing and Infrastructure (DPHI) to support the State-led rezoning of the Macquarie Coal Complex for Post Mining Land Use. The Macquarie Coal Complex Land Transformation Precinct (the Precinct) forms part of the NSW Government's investigation into future land use pathways for mining affected land.

The former operations of the Macquarie Coal Complex have created a legacy Tailings Storage Facility (TSF), an engineered structure designed to safely store and manage rejected materials generated from operations, including solids and water. Closure of the TSF, including rehabilitation and capping is underway however will remain a declared dam until water has been removed from the contained material and the structure is stabilised.

Dambreak modelling has defined inundation extents, flow paths, and level of hazard under a failure scenario. These outputs directly inform spatial planning by establishing compatible land use scenarios across the TSF areas that would enable flexibility over the period of rehabilitation and closure. Sensitive or high-occupancy land uses are avoided in the short to medium term within mapped inundation zones and buffers. Lower-intensity or transient uses may be considered in the outer zones (such as passive and active recreation, storage facilities), subject to appropriate risk mitigation measures. Areas above the TSF could provide land uses which do not compromise the capping once rehabilitation is complete and a more stable landform is achieved over time.

New and existing infrastructure, such as roads that traverse the dambreak inundation area and have the potential to increase societal risk, must be appropriately designed to ensure they are located above the modelled dambreak inundation level whilst the TSF is a declared dam.

A strategic, risk-informed framework to manage constraints associated with the TSF and the potential for dambreak downstream (dambreak inundation area) should therefore be adopted. This ensures that future land use is compatible with the residual risk profile of the TSF over its remaining life and following de-declaration.

1.1 Background

The Macquarie Coal Complex is a decommissioned mining operation previously operated by Glencore, which processed coal sourced from the nearby underground West Wallsend Colliery and the Westside open-cut mining operations. Mining activities at West Wallsend Colliery ceased in 2016.

The Tailings Storage Facility (TSF), commissioned in 1983, has been the sole tailings storage facility at the site. The purpose of the TSF is to provide storage for the disposal of fine reject material (i.e., tailings). The tailings were historically pumped in slurry form to the TSF. Coarse rejects were trucked to the Rejects Emplacement Area (REA) situated immediately adjacent to the TSF, on the eastern side.

While infrastructure rehabilitation has been ongoing, the TSF area has remained largely dormant since closure to allow for the drainage and consolidation of the tailings. The only activity undertaken during this period has been the placement of fine ash tailings to facilitate site access. Works to achieve the final landform of the TSF, including geotechnical stability are currently underway. Capping will be the final stage of rehabilitation of the TSF.

1.2 Purpose of this report

The purpose of this Dam Engineering Report is to:

- Analyse the available information on the existing Tailings Storage Facility (TSF) to better understand the annual probability of failure and the dam safety risks associated with the TSF, in particular the northern saddle dam (NSD)

- Provide a framework to support inputs into the *State Environmental Planning Policy (Precincts-Regional) 2021*
- Provide a rezoning appropriate evidence base to verify developable land and inform land use compatibility, built form intensity and infrastructure feasibility
- Provide developable land and constraints classification to support the Structure Plan
- Translate constraints, including tailings storage facility (TSF) and reject emplacement limitations, steep terrain constraints, and mine subsidence district considerations into Master Plan mapping and performance criteria
- Provide screening-level land capability advice
- Identify landform trajectory considerations where development potential may improve over time (e.g. post-settlement or completion of stabilisation activities), and translate this to 'short/medium/long-term' triggers
- Provide a clear pathway for post rezoning phase investigations and considerations (further investigations, testing, monitoring etc).

1.3 Methodology

The methodology adopted for this Report is desktop based and summarised as follows:

- Review of available information on the TSF regarding the investigation, design and construction
- Review of dambreak & consequence assessment
- Review of surveillance and monitoring practice implemented at the dam
- Review of dam safety risks
- Review of land use planning scenarios within the Precinct
- Analysis of activities required to inform post mining land use planning as part of a State led rezoning.

1.4 Reliance on data and information

The advice and recommendations in this Report rely on the accuracy of the existing input documentation and advice provided to date. This includes the following documentation:

1. Post Mining Land Use – Deliverability of Pilot Projects – Due Diligence Report – Macquarie Coal Complex, prepared by Aurecon in February 2026
2. Insights obtained from internal workshops with DPHI, package leads and agencies
3. Dam break failure impact assessment (ATCW, 2023)
4. Surveillance review reports (ATCW 2021, 2022, 2023, 2024, 2025)
5. West Wallsend Dambreak (Presentation Slides) (2024)
6. Design Record Report (ATCW, 2025)
7. TSF risk assessment and SFAIRP framework report (ATCW, 2025)
8. West Wallsend Tailings Dam Landform Finalisation (WSP, 2025).

This Report has also been informed by a three-day Enquiry by Design (EbD) workshop held with DPHI, the NSW Resources Regulator, Glencore and other key stakeholders held in May 2026.

No detailed site investigations or modelling has been undertaken to inform the analysis or recommendations in this Report.

2 Site location and context

Located within the Lake Macquarie Local Government Area, the Precinct comprises approximately 1160 hectares of land across the suburbs of Killingworth, Wakefield, Barnsley and Teralba. The Precinct forms part of the broader Lower Hunter region and is strategically positioned within an established industrial and infrastructure corridor, with proximity to key regional centres (Figure 2-1).

Macquarie Coal Complex is a former mining operation comprising West Wallsend Colliery (underground mine), Westside Mine (open cut), the Macquarie Coal Preparation Plant (MCP), Tailings Storage Facility (TSF) and associated infrastructure. Coal mining has been a defining land use in the area since operations commenced in 1969, shaping both the physical landscape and the economic profile of surrounding communities. The Macquarie Coal Complex site is currently closed and undergoing rehabilitation.



Figure 2-1 Broader Precinct context

3 Regulatory Context

In New South Wales (NSW), dam safety is enforced by the *Dams Safety NSW Act 2015* (the Act) and the *Dams Safety NSW Regulation 2019* (the Regulation). Dams Safety NSW is the state regulator for dam safety and responsible for developing and implementing regulations for effective dam safety management to protect life, property, and the environment from dam failures. Dam Safety NSW replaces the former Dams Safety Committee (DSC).

Owners of declared dams, and their professional advisors, are accountable to determine and take appropriate dam safety management action to prevent the potential failure of their dams and to mitigate associated consequences. Responsibility for dam safety obligations remains with the declared dam owner unless formally transferred in accordance with the Act, requiring regulatory approval. Dams Safety NSW will audit declared dam owners' policies, processes, and procedures to check compliance with the requirements of the Regulation.

In accordance with the *Regulation*, declared dam owners must establish, implement and maintain a Dam Safety Management System (DSMS) to ensure the ongoing safety of their dam throughout the dam's lifecycle. The DSMS provides a comprehensive and integrated system for the management of all aspects of foreseeable risks in relation to the failure of the dam. It provides a structured approach for the development, coordination and control of activities undertaken to ensure that these dam failure risks are reduced so far as is reasonably practicable (SFAIRP). The DSMS must address the following as set out in the *Regulation*:

- a. *the procedures for managing risks in accordance with Clause 14,*
- b. *the procedures for reporting incidents in accordance with Clause 19,*
- c. *the procedures for conducting a safety review in accordance with Clause 20,*
- d. *the procedures for notifying changes to the dam in accordance with Clause 21,*
- e. *the procedures for ensuring compliance with AS ISO 55001:2014,*
- f. *the procedures for review of the matters in paragraphs (a) to (e),*
- g. *the resources that will be applied for the effective implementation and use of the DSMS.*

In accordance with the *Regulation*, *Clause 14*, the owner of a declared dam must establish a risk management framework inclusive of the following components:

- hazard identification process
- a risk analysis process
- a risk evaluation process
- a risk treatment process.

Risk in dam safety context is the combination of the probability of dam failure and the resulting consequences. The risks posed by a dam involves societal risk, individual risk and financial/economic risk. Dam Safety NSW has provided thresholds for societal risk and individual risk as follows:

- Societal risk rating must be lower than 1E-03
- Individual risk rating must be lower than 1E-04.

Societal risk is also presented in societal risk plot, also called F-N plot. Societal safety threshold specified by Dam Safety NSW (2022) for an existing dam is presented in Figure 3-1. The societal risk posed by the dam must be lower than the safety threshold and so far as is reasonably practicable (SFAIRP).

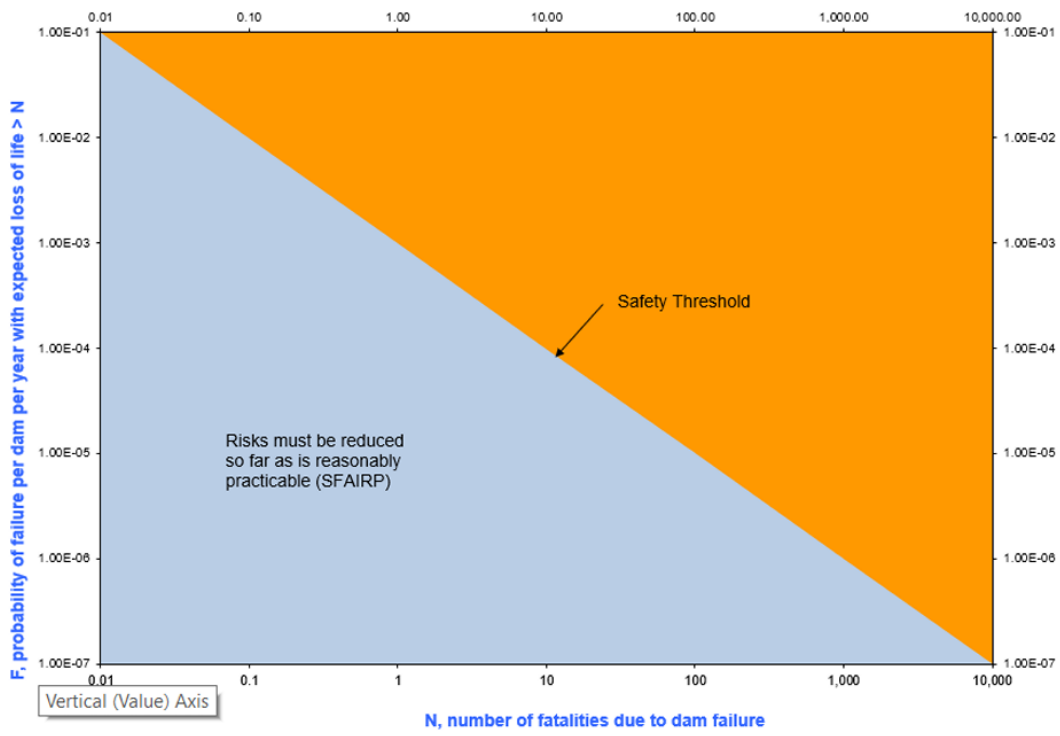


Figure 3-1 Societal safety threshold for existing dams

4 Baseline analysis

4.1 Investigation, design and construction of TSF

The surveillance report (ATCW, 2024) reports that geotechnical investigations at the TSF were conducted in stages. The Phase One Study comprised of a review of existing conditions, embankments, in-situ tailings characteristics, capping concept development, and an initial assessment of tailings consolidation to assist with final landform design.

The Phase Two Study was completed in January 2018 and focussed on obtaining additional geotechnical data. This included CPTu testing, drilling, sampling, and laboratory testing of the tailings and embankment coarse reject materials. It is not clear if the investigations were also conducted at the NSD and eastern embankment to establish the material parameters for these embankments and the underlying foundations.

The Phase Three Study addressed further works, specifically related to the Rejects Emplacement Area (REA), that was recommended by ATCW resulting from the Phase Two Study, as well as recommendations by a third-party reviewer.

Tailings

The fine-grained tailings exhibit compressibility and permeability values ranging from 0.06 to 0.4 m²/year, indicating poor consolidation characteristics. Laboratory testing of the clay fraction confirms mineralogy dominated by illite, swelling smectite, and kaolin. The in-situ strength profile of the underlying tailings (including the capping layer) is not currently available and will require reassessment as part of final landform design. For post-seismic assessment, a residual undrained shear strength ratio ($S_{u,res}/\sigma'_{vo}$) of 0.04 has been adopted for the tailings and fly ash. This value corresponds to the 20th percentile of strengths derived from CPT data within the weakest tailings zone adjacent to a hypothetical breach location.

Embankment Materials

The embankment materials comprise approximately 85% cobble- and gravel-sized particles, with the remaining 15% consisting of sand and fines (silt and clay). For analysis, the embankment has been subdivided into 'new' and 'old' rejects, with drained (effective) strength parameters applied to the new rejects and undrained strength parameters applied to the old rejects. Although laboratory testing indicates broadly similar particle size distribution and plasticity for both materials, undrained conditions are not considered appropriate for the new rejects, as these have historically remained unsaturated.

The drained shear strength for the REA "new rejects" indicates an effective friction angle (ϕ') of approximately 38°, assuming zero cohesion. The undrained shear strength ratios range from 0.35 to 0.40 for REA rejects and from 0.50 to 0.80 for old rejects. CPTu data indicate that the minimum undrained shear strength of the coarse rejects consistently exceeds 20 kPa.

Following completion of the final landform, the phreatic surface within the embankment is expected to gradually decline. Seepage modelling undertaken by ATCW in 2020 indicates that the phreatic surface will remain within the embankment for approximately 15 years following closure.

A site-specific seismic hazard assessment completed in February 2017 adopted peak ground accelerations of 0.19 g, 0.25 g, 0.43 g, and 0.53 g for return periods of 500, 1,000, 5,000, and 10,000 years respectively. Based on ATCW (2024), as part of these studies, the safety, in relation to earthquake loading on the NSD in its current condition, was examined and it was concluded that the deformations resulting from the Maximum Design Earthquake (now Safety Evaluation Earthquake) would be tolerable and would not result in the uncontrolled release of tailings or water from the TSF.

The current risk assessment reflects the status of capping progress and tailings deposition levels as of October 2025. The dam's risk profile is expected to change following completion of capping, particularly as the primary risk driver—overtopping associated with scarp formation and slumping at the buttress toe leading to global slope instability (94.68%)—will be reduced over time as stormwater will be diverted away from the TSF and there will be natural infiltration.

4.2 Dambreak & consequence assessments

This section provides a strategic, desktop-based assessment of existing dam break conditions, constraints and opportunities associated with the Tailings Storage Facility (TSF) within the Precinct. It also provides recommended actions for the post-rezoning phase. The assessment considers the TSF embankments, capping works, potential dam failures, previous dam break assessments and dam safety requirements, and included a review of the following documents:

- Dam break failure impact assessment (ATCW, 2023)
- West Wallsend Dambreak Presentation Slides (ATCW, 2024)
- Surveillance review reports (ATCW 2021, 2022, 2023, 2024, 2025).

The TSF has two embankments: the eastern embankment (EE) and northern saddle dam (NSD). Following the completion of Stage 1 capping, Stage 2 capping commenced in 2021. Thus far, for Stage 2 capping (29 June 2021 to November 2024), approximately 435,000 m³ (~356,000 m³ in-situ after initial settlement) of fly ash has been hydraulically deposited into the TSF. Stage 2 Phase 3 fly ash deposition has been suspended to allow conventional capping.

Of the two embankments, the eastern embankment has a wide crest and flat downstream slopes as a result of the Rejects Emplacement Area (REA) progressively constructed by dumping, spreading and track-rolling coarse rejects in lifts to an ultimate surface level to that of the eastern embankment. As such, only the failure of the NSD is envisaged.

The latest dambreak study was undertaken in 2023 (ATCW), which considered three incremental failure consequences for the NSD. The population at risk (PAR) was updated in 2024 and potential loss of life (PLL) was also estimated in 2024. Incremental Flood Failure (IFF) considered in the dambreak of NSD in 2023 is very conservative. A scenario for spillway in operation without obstruction and with downstream concurrent inflow during PMF (natural flooding) without dambreak compared with dambreak will be more realistic for IFF.

The dam was re-classified in 2024 as 'Significant' consequence category dam for sunny day failure based on a PLL of 0.6 and a medium severity of damage and loss. The surveillance report (ATCW, 2024) and the presentation slides of the updated consequence assessment do not provide information on flood failure related PAR/PLL. TSF is a declared dam under the Dam Safety Act 2015 and the *Dam Safety Regulation 2019*. TSF must comply with the requirements of the *Dam Safety Act 2015* and the *Dam Safety Regulation 2019*.

Inundation extent during a sunny day failure as shown in the surveillance report (ATCW, 2024) is presented below in Figure 4-1.

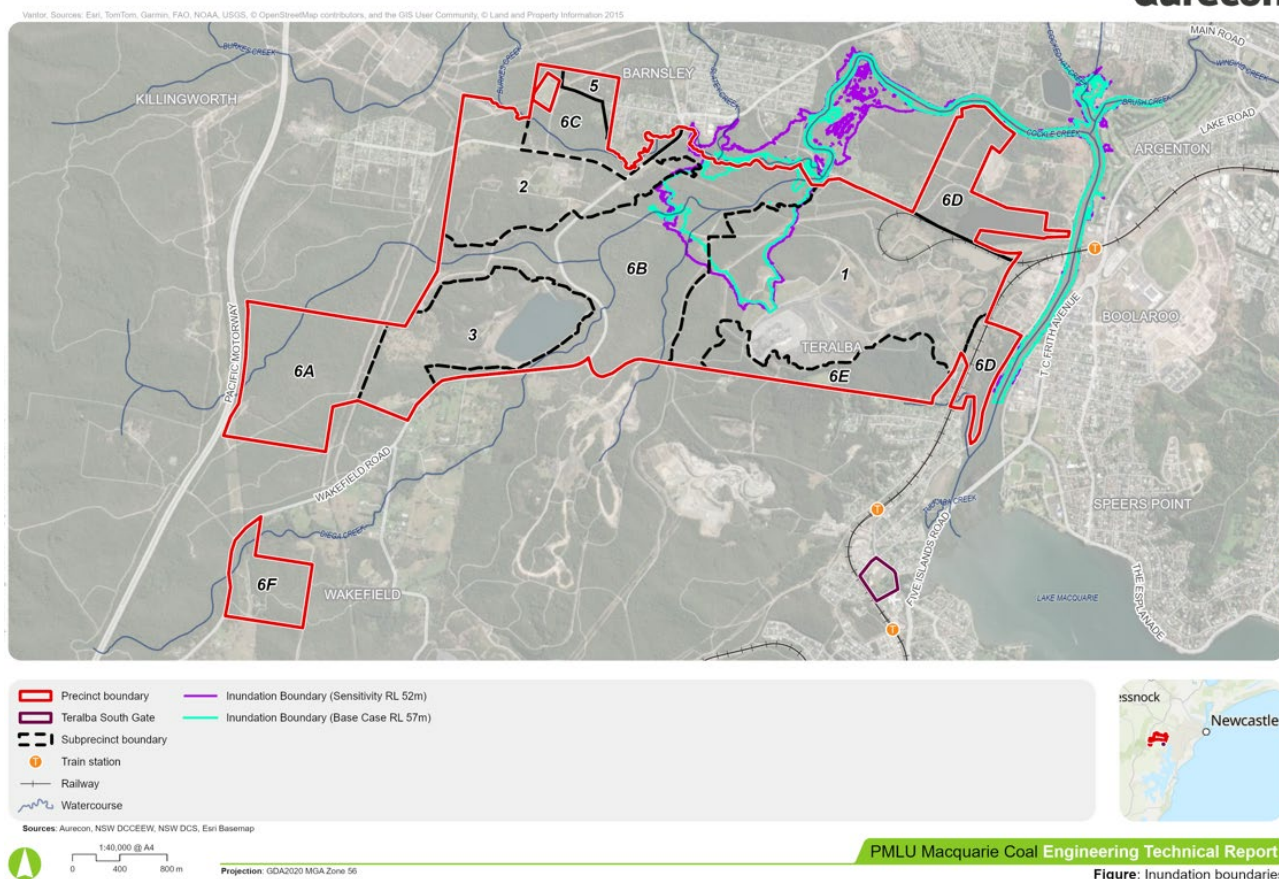


Figure 4-1 Dambreak inundation extent of a sunny day failure

4.3 Surveillance and monitoring practice implemented at the dam

The owner of the dam (NSD and TSF) is responsible for operating and maintaining the dam in a safe manner until such time that the dam is de-declared, which requires undertaking routine monitoring activities, periodic reviews and interpretation of collected data, dam safety surveillance inspections (intermediate, comprehensive) and special inspections following an unusual loading at the dam (e.g. flood or seismic). The *Dam Safety Act 2015* and the Regulation (2019) require that a Dam Safety Management System (DSMS) is prepared for all declared dams, including Operation and Maintenance Plan (OMP) and Dam Safety Emergency Plan (DSEP).

Review of previous surveillance and monitoring reports for the TSF reveals that:

- The TSF has established a Dam Safety Management Plan (DSMS) in 2023
- There is an Operation and Management Plan for the TSF
- A Dam Safety Emergency Plan (DSEP) exists for the dam which defines white, amber and red Trigger Alert Response Plans (TARPs)
- The performance of the dam (NSD) is mentioned to be satisfactory per the surveillance report by ATCW (2024). It reports a 10-metre-wide scarp and slump associated with it near the downstream toe caused by incessant rainfall. This has since been remediated. Minor issues on the capping have been continuously monitored and repaired as required
- The monitoring and surveillance requirements are largely based on the frequencies recommended by ANCOLD (2003) or on a risk-based surveillance (RBS) basis. The surveillance report (ATCW, 2024) mentions that currently routine visual inspection is undertaken at the dam on a weekly basis and are considered to be in compliance with the requirements of ANCOLD (2003) and Dam Safety NSW

- Intermediate surveillance inspections and reporting for the dam are undertaken on an annual basis by Engineer of Record. Five yearly comprehensive surveillance inspection and reporting are undertaken once every five years by an Independent Dam Specialist
- Given that the risk assessment was undertaken recently in 2025, the next 5-yearly risk review is due in 2030
- The TSF has following instrumentation currently for monitoring of the dam performance:
 - There are ten (10) active standpipe piezometers with real time sensors installed within the NSD which extend into the foundation
 - A v-notch weir with automated seepage monitoring installed together with a camera
 - Seven (7) monitoring bores were installed within the Reject Emplacement Area (REA) to establish an understanding of phreatic surface levels in the area. REA is situated immediately adjacent to the TSF, on the eastern side
 - A v-notch weir also exists in the south-east REA toe drain adjacent to the TSF South Ramp
 - In November 2023, a ground control network to facilitate aerial (photogrammetry) surveys was installed. The network consists of thirty-two (32) ground control points and ten (10) check control points. Twenty-five (25) survey prisms were also installed across the downstream slope of the NSD to monitor embankment movements.
- The NSD standpipe piezometers and v-notch weir are continuously monitored using eagle.io (with assigned trigger levels). Embankment movement is monitored fortnightly using birdi.io (with assigned trigger levels). Additionally, storage and ongoing fly ash deposition are monitored regularly via photogrammetry processed in the birdi.io platform. The collected data is analysed weekly to understand the developing trends.

4.4 Dam safety risk

The following information is gathered from the NSD risk assessment and SFAIRP framework report by ATCW (2025).

Engineering assessment for the dam (embankment and spillway) was undertaken in 2025 following engineering standards, guidelines and references.

Risk assessment was then undertaken in a sequence starting with failure mode and effect analysis (FMEA), load partitioning (flood, seismic), event trees (piping and non-piping), estimation of conditional probability of failure, consequence of failure, risk integration (annual probability of failure, risk estimates) and risk evaluation.

The annual probability of failure for the dam was estimated to be $1.74\text{E-}07$. The most significant contributor was overtopping due to scarp formation and slump at the buttress toe leading to global slope instability, contributing more than 94% of the overall annual probabilities.

The total societal risk was estimated to be $5.02\text{E-}08$. It is not clear what the societal risk rating was, yet it would still be lower than the total societal risk, the threshold for which is $1\text{E-}03$. The societal risk (F-N) plots four orders of magnitude below the Dam Safety NSW (2022) safety threshold.

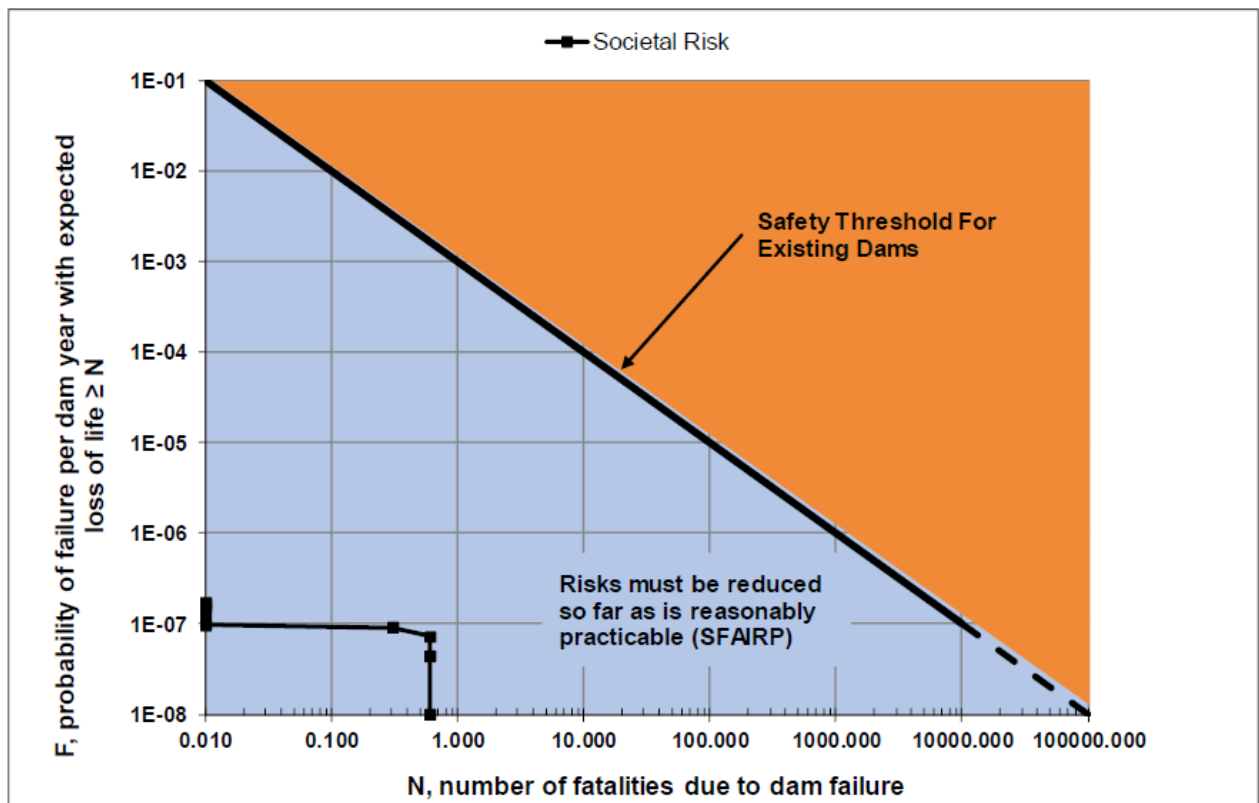


Figure 4-2 NSD societal risk plot (ATCW, 2025)

The total individual risk posed by the dam was estimated to be 3.22E-12. The highest individual risk is not clear in the report, yet it will still be lower than the total. The Dam Safety NSW (2022) threshold for individual risk rating is 1E-04.

It should be noted that the safety threshold by Dam Safety NSW (Gazette 113, 2022) states that even if the risk plots below the threshold, the risks must be reduced so far as is reasonably practicable (SFAIRP). An SFAIRP assessment was undertaken following the risk assessment for NSD in 2024. The outcomes of the SFAIRP assessment indicated that the existing control measures were consistent with recognised good industry practice. No additional control measures were identified.

The risk assessment reflects the current condition of the structure, with capping progress and tailings deposition levels recorded as of October 2025. It is noted that the post-capping landform is expected to alter the risk profile of NSD; however, this has not been considered in the current assessment.

It should be noted that any development activity implemented downstream of the dam within the current inundation area will most likely increase the PAR in the form of itinerant population as well as permanent population. In that case, it is highly likely that the PLL will increase as well.

While the probability under the post-landform condition is expected to decrease, future development on and downstream of the dam may increase the Potential Loss of Life (PLL), which is contrary to the Dam Safety NSW (2022) directive. Accordingly, the overall societal risk must remain below the prescribed safety threshold to maintain compliance. The societal risk estimated in the latest assessment is more than four orders of magnitude below the DSNSW safety threshold, and this margin should be maintained under post-landform conditions.

5 Structure Plan Assessment

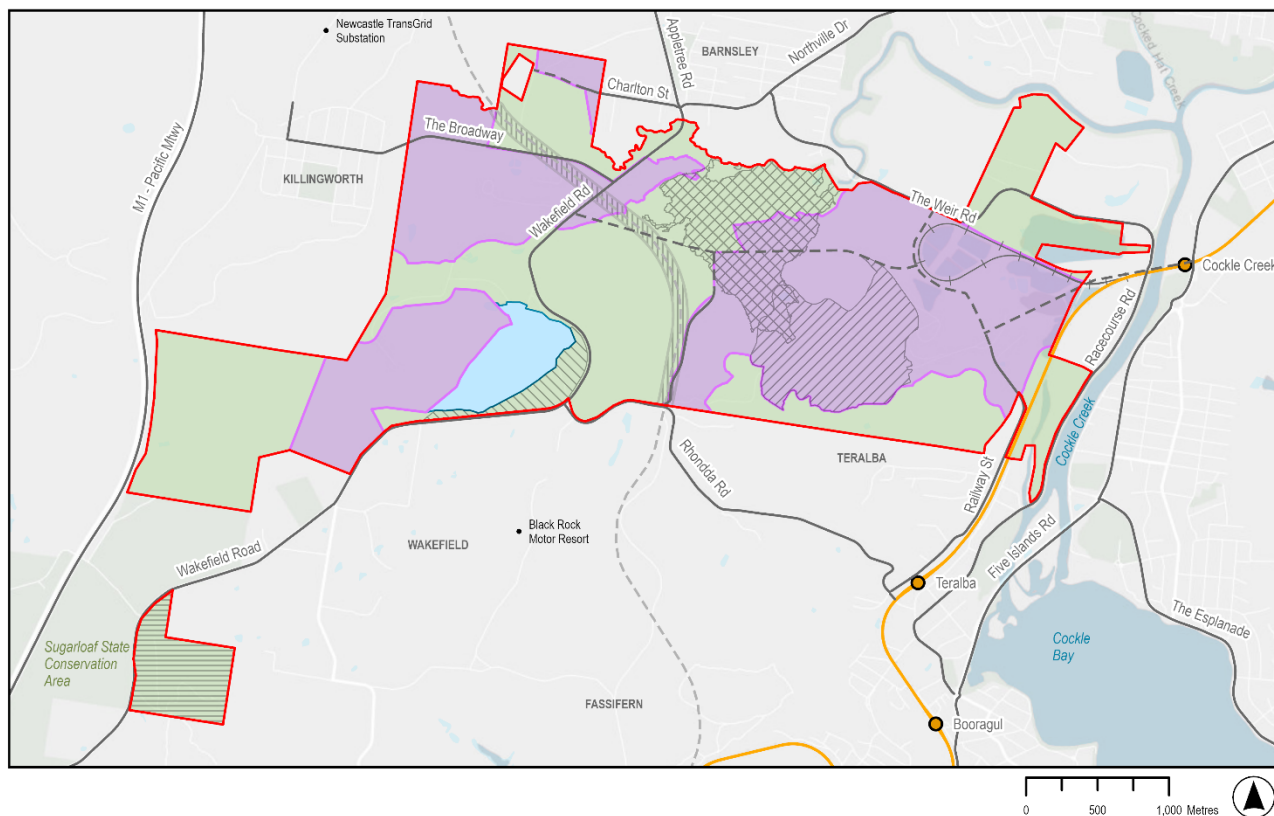
The Macquarie Coal Complex Land Transformation Precinct Structure Plan (Structure Plan), is the principal spatial framework underpinning the Master Plan (Figure 5-1). It provides developable areas, environmentally constrained areas, recreation areas, movement networks and key constraints across the Precinct.

Categories relevant to this Report are as follows:

- Developable areas – Land assessed as most likely to be suitable for development, subject to further technical investigation and site-specific assessment at the development application stage. Developable areas are concentrated in locations with relative geotechnical stability, accessibility to existing or planned transport networks and proximity to utility infrastructure or feasible servicing solutions
- Environmentally constrained areas – Land identified as having high biodiversity values or where natural features (watercourses, wetlands, steep slope, flooding) constrain development for employment uses. Detailed environmental investigations are required prior to development or disturbance of vegetation in these areas. Environmentally constrained areas form an interconnected network across the Precinct that underpins ecological connectivity, water cycle management and landscape identity
- Recreational areas – Land suitable for active and passive recreation facilities, balanced with protecting biodiversity and cultural values. Recreational areas are primarily associated with the West Lake sub-precinct and the environmentally constrained areas, where the landscape and environmental values support nature-based tourism, walking and cycling trails and community recreational facilities
- Developable land subject to further investigation – Land constrained in the short to medium term by geotechnical, dam safety, flood or biodiversity factors requiring resolution before development can proceed. Further investigation is required, as described in this Master Plan, prior to development. The release of this land for development is contingent upon the completion of relevant technical investigations and the achievement of applicable staging triggers
- Movement network – Existing and proposed road, freight and active transport corridors that structure internal and external connectivity
- Key constraints – Significant infrastructure, environmental or hazard features (e.g. TSF, pipeline, coastal wetlands, transmission lines) that influence land use allocation and development design.

At the time of reporting, the TSF remains a declared dam. Subsequently, development within the dambreak inundation zone and above the TSF itself is constrained until the TSF is de-declared. The potential of developable land within and adjacent to the inundation zone is contingent upon updated dam break and consequence assessments demonstrating a reduction in risk as rehabilitation milestones are achieved.

The Master Plan provides overarching performance criteria to manage dam risk and support progressive development over time.



Legend

- | | |
|--|--|
| Precinct boundary | Environmentally constrained areas |
| ● Railway station | Developable areas |
| —+— Rail loop | Recreation areas |
| — Railway line | Further investigation area - dam break inundation zone |
| — Road | Further investigation area - tailings storage facility |
| - - - Road - potential | Further investigation area - West Lake |
| - - - Indicative Lower Hunter Freight Corridor | Further investigation area - Wakefield |
| | Further investigation area - Lower Hunter Freight Corridor |

Figure 5-1 Macquarie Coal Complex Land Transformation Precinct Structure Plan (Master Plan, GHD 2026)

The Structure Plan is organised into five sub-precincts (Figure 5-2). The TSF is located within the Cockle Creek sub-precinct, to the eastern portion of the Precinct.

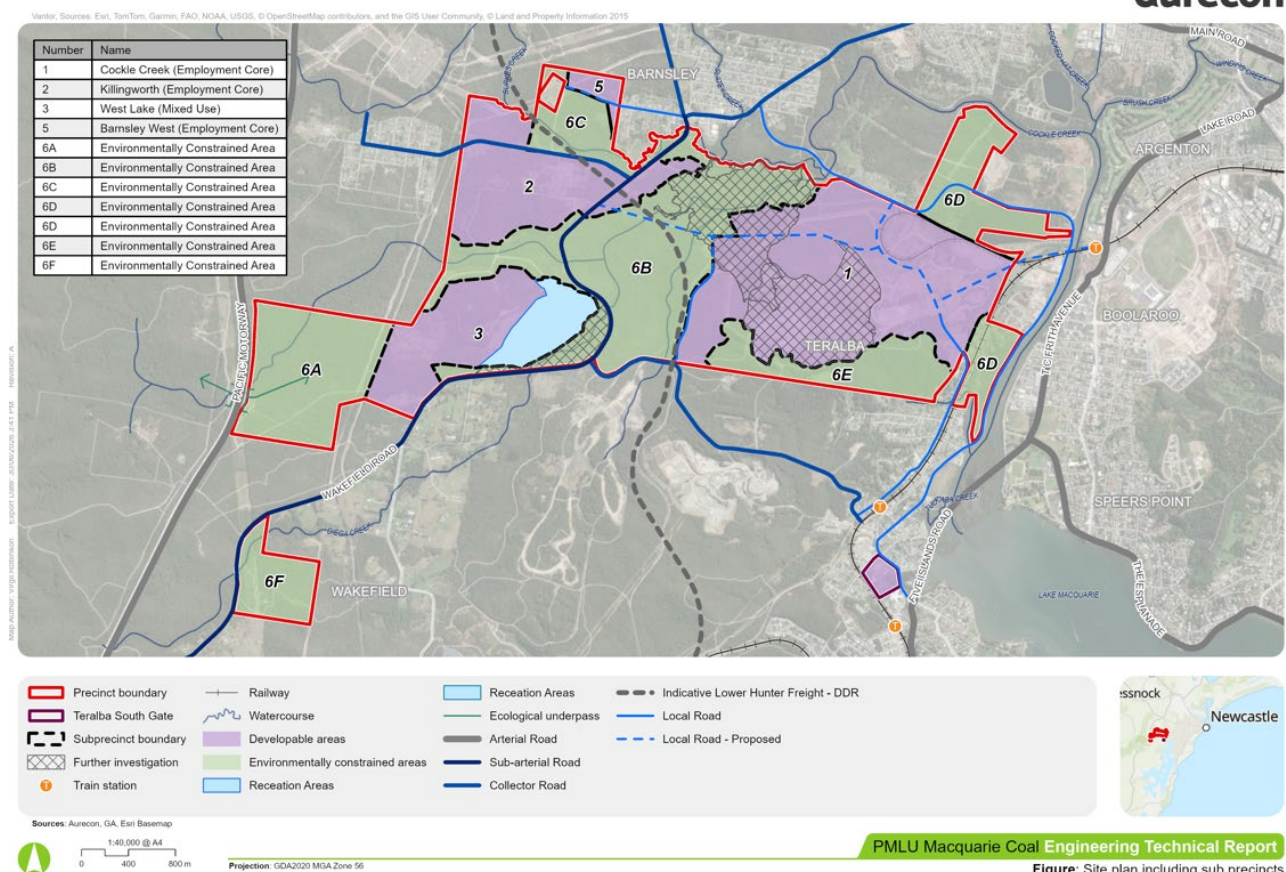


Figure 5-2 Sub-precincts (Master Plan, GHD 2026)

Cockle Creek sub-precinct seeks to accommodate a range of employment-generating uses, including general industrial, manufacturing, logistics, warehousing and freight-related uses that leverage the sub-precinct's central location, existing industrial infrastructure and proximity to the regional road network. The former mine rail loop and rail interface present a potential long-term opportunity for low-frequency rail freight, subject to feasibility assessment.

The TSF and associated dambreak inundation zone represent the most significant constraint to near-term development (Figure 5-3 and Figure 5-4). Early activation is limited to areas outside the TSF and inundation area, with the progressive release of developable land contingent upon TSF rehabilitation and potential future de-declaration.



Figure 5-3 Cockle Creek sub-precinct aerial map showing the TSF (Master Plan, GHD 2026)

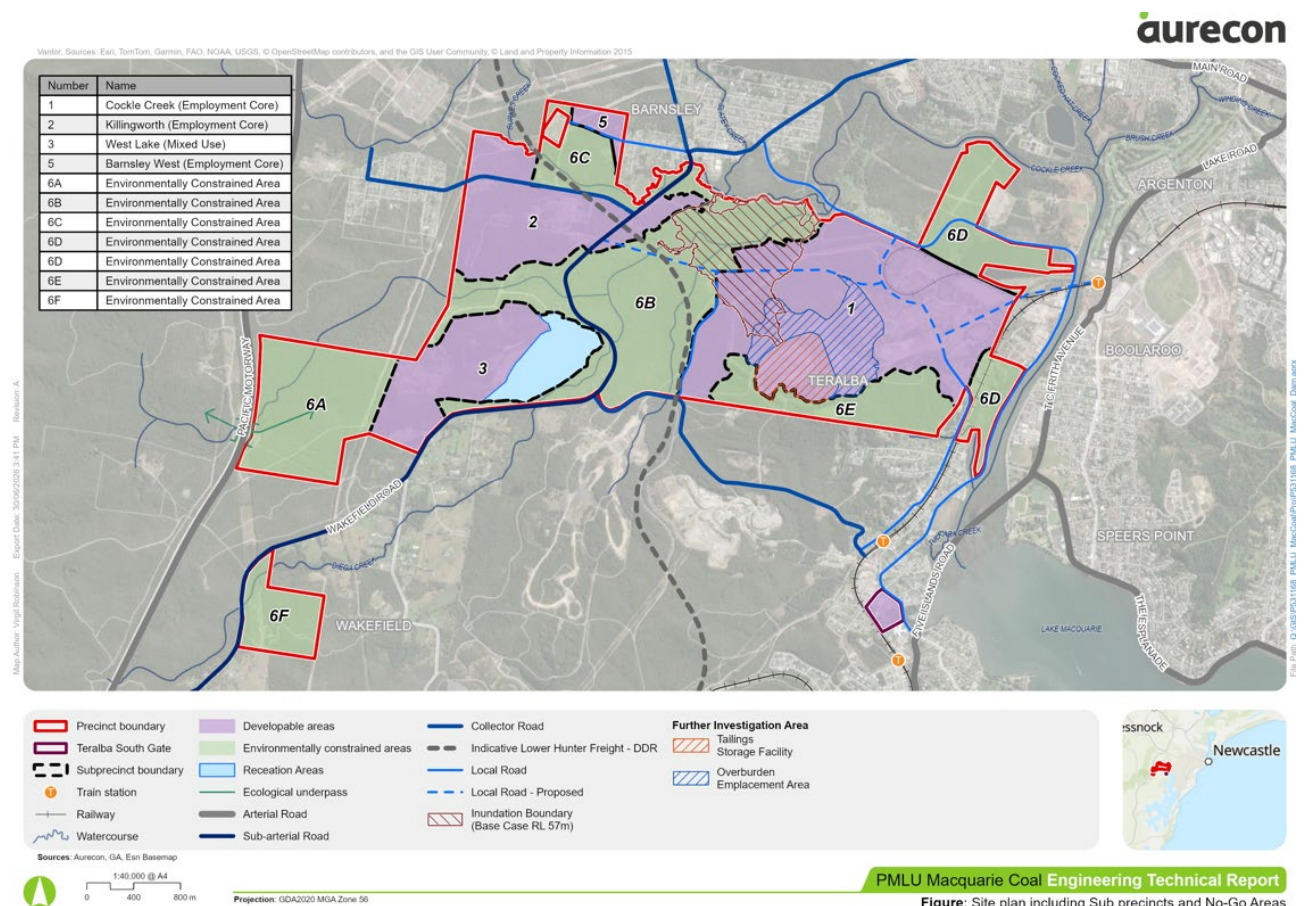


Figure 5-4 Structure plan identifying the zones of the TSF

5.1 Downstream dambreak inundation zone

5.1.1 Dambreak and consequence assessment

The update to population at risk (PAR) and potential loss of life (PLL) undertaken in 2024 by ATCW considers sunny day failure related PAR and PLL. Flood failure related incremental PAR and PLL is calculated based on the dambreak inundation extent for PMF events with and without dam failure. The inundation extent for a flood failure may be different to a sunny day failure inundation extent, although the incremental PAR/PLL may be insignificant.

Currently, development is restricted to areas higher than the maximum inundation levels on the downstream side of the dam as it will have potential to increase the PAR/PLL which may increase the baseline dam safety risks in contrast to the requirement of Dam Safety NSW (Gazette 113, 2022), which directs that the risks must be reduced so far as is reasonably practicable (SFAIRP).

5.1.2 Rehabilitation

The areas downstream of the dam that are impacted by natural flooding should be rehabilitated after each flooding event. This might include maintaining the walking tracks, erosion protection works, routine vegetation management along the floodplain, construction of flood levees to protect areas of significance from the next flooding etc. Severe change to the inundation zone may have significant impact on the dambreak inundation flow characteristics (flow depths, velocities and extent) compared to what has been estimated in the 2023/2024 dambreak modelling.

5.1.3 Development within the inundation zone

The area downstream of the NSD may be suitable for the development of walking trails, subject to appropriate risk management constraints. It may also be utilised for low-impact biodiversity enhancement works, provided that such activities do not obstruct potential dam break flood pathways or increase inundation levels.

Following the completion of TSF rehabilitation works, and as the phreatic surface within the TSF progressively lowers to a stable condition, updated dam break assessments should be undertaken periodically. These assessments will refine the predicted inundation extents over time.

As inundation extents reduce, portions of land downstream may be progressively released for potential development, subject to confirmation of geotechnical stability and safety constraints. Concurrently, land-use option studies can be carried out in alignment with the ongoing dam management program, while the dam remains declared

Furthermore, infrastructure such as roads which traverse the dambreak inundation zone and increase societal risk will need to be raised above the modelled inundation level and designed to withstand flood impacts whilst the dam remains declared.

5.2 Northern Saddle Dam (NSD)

5.2.1 Dam safety surveillance and monitoring

Based on the review of the surveillance reports, monitoring and surveillance at the dam shall continue following ANCOLD (2003) for a Significant consequence category dam. The evaluation of instrumentation data should be undertaken at least quarterly to understand the performance of the dam and early detection of any developing trends.

5.2.2 Dam safety review and risk assessment

Dam safety reviews are generally undertaken in 10-to-15-year intervals, unless earlier review is warranted due to change in loading conditions or deterioration in the condition of the dam. These periodic reviews benefit from the availability of additional information over time, including outcomes of investigations and studies, as well as updates to hydrological and seismic data. They also take into account any advancements in engineering standards, guidelines, and industry best practices. The findings of each dam safety review inform updates to the dam's risk assessment. Following this, a so far as is reasonably practicable (SFAIRP) assessment should be carried out to evaluate whether further risk reduction measures are justified and proportionate.

Dam Safety NSW (2022) requires that the risk is reviewed once every 5 years. As such, at the time of reporting, the 5-yearly risk review is due in 2030.

5.2.3 Upgrade to the dam

Dam safety risks are fundamentally associated with the potential for dam failure. These risks can only be eliminated if all credible failure modes are demonstrably removed, such that no plausible failure scenario can be reasonably envisaged. As an example, the TSF is impounded by the NSD and Eastern Embankment. The Eastern Embankment has been made significantly wider at crest and flat downstream slopes as a result of the Rejects Emplacement Area (REA), progressively constructed by dumping, spreading and track-rolling coarse rejects in lifts to an ultimate surface level to that of the eastern embankment. As such, only the failure of the NSD is envisaged. If similar buttressing to the NSD can be done with engineered fill sufficiently wide enough with flatter slopes, that would eliminate the global failure of the NSD. Practicality of this needs to be investigated as this might reduce the duration for de-declaration.

5.2.4 Development at the NSD

The TSF embankment requires ongoing routine monitoring to identify any developing conditions that may compromise dam safety. This monitoring is carried out through regular visual inspections by trained personnel. Accordingly, key structural components of the embankment, including the upstream and downstream faces, crest, groins (both upstream and downstream on the left and right sides), abutments, and the embankment toe, collectively define the dam footprint and must be preserved. These areas must not be altered, encroached upon, or used for any development activities, as unrestricted access is required for inspection, surveillance, and maintenance purposes.

The dam footprint is also subject to periodic preventative and corrective maintenance. To minimise the risk of internal erosion (piping), no vegetation should be permitted to grow, or be planted, within a 10 metre buffer zone surrounding the dam footprint. Vegetation root systems can create preferential seepage pathways, potentially compromising the structural integrity of the embankment.

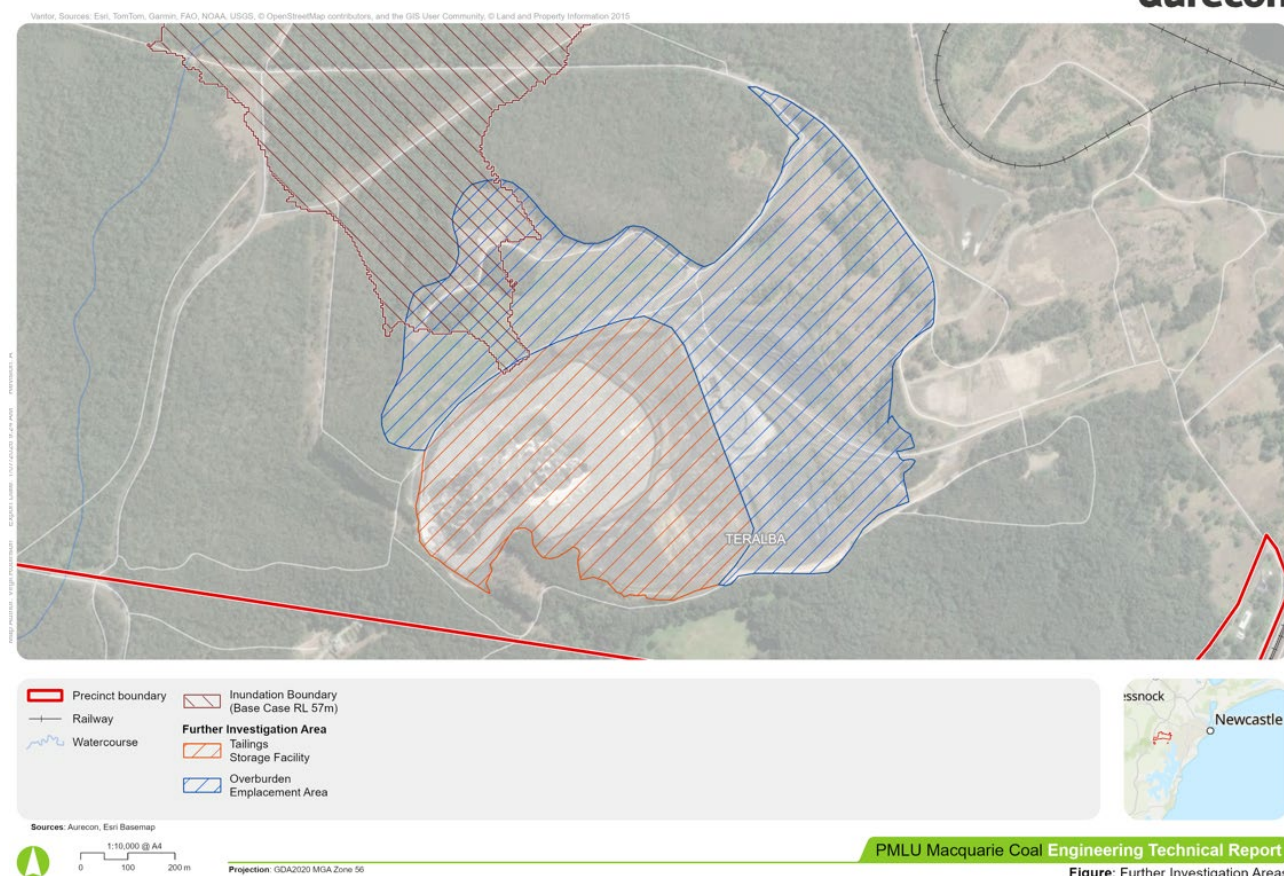


Figure 5-5 Further investigation areas - TSF and reject emplacement area

5.3 Tailings Storage Facility (TSF)

5.3.1 Investigations, monitoring and surveillance

The existing seepage modelling is considered conservative due to uncertainties in some of the adopted strength parameters. This can be refined through the installation of additional groundwater monitoring wells around the tailings storage to better define current water levels and flow directions, as well as through targeted testing to improve parameter confidence.

5.3.2 Maintenance of final landform

Once the final landform is established, the tailings material will continue to consolidate, and the capping will continue to settle over time. Some ongoing work to rehabilitate impacted areas will be required to maintain the designed levels and ensure that the area does not pose safety hazards due to factors such as uneven terrain, localised erosion and ponding.

5.3.3 Development at the TSF

On the TSF, the rehabilitated area can be used for recreational facilities; however, due to non-uniform settlement over time the surface may become uneven. This will limit the development on the TSF. Subject to risk criteria and geotechnical constraints, some low-risk land uses can be considered, such as walking trails and passive recreation. Plantation on the TSF shall be avoided. Option studies for land use can be undertaken with ongoing dam management programme if the dam remains declared. Once the dam is de-declared, there may be an opportunity to reuse tailings for industrial use. Whilst the TSF may be de-declared there remains an ever-present risk that future development works on the TSF may adversely disrupt the stability of the TSF (such as through reintroduction of water to the TSF material). This will also include that any investigatory works and ensuring the site is returned to a secure safe state.

5.4 De-declaration of NSD

5.4.1 Natural infiltration

Currently, the rehabilitation of the TSF is being undertaken, and a final landform will be created which will allow channelling run-off from the TSF area to drains on the finished impermeable surface and route the run-off away from the TSF. This will prevent rainfall infiltration into the TSF. The stored tailings will rely on natural infiltration for reduction in the phreatic level. Once the TSF does not have significant volume that can flow downstream during a potential dambreak event and that there is no PAR/PLL, the dam will be a candidate for de-declaration. Even then, the inundation extent (much reduced compared to the baseline extent) shall be avoided for any development that will have the potential to introduce PAR/PLL.

5.4.2 Accelerated dewatering

The de-declaration process may be accelerated by providing additional drainage pathways within the tailings, such as through the installation of horizontal or vertical drains. The effectiveness of these measures should be assessed during the concept design stage, particularly given the relatively low permeability of the tailings.

The current water level within the tailings is not well defined; therefore, installation of additional groundwater monitoring wells is recommended to better characterise groundwater levels and flow directions.

5.5 Conclusions and recommendations

The TSF is a declared dam under the *Dam Safety Act 2015* and *Dam Safety Regulation 2019*. This requires complying with the Dam Safety legislation to ensure that the dam is operated and maintained safely, required assessments are undertaken from time to time and implementation of remedial or safety measures mandated by the authority are undertaken.

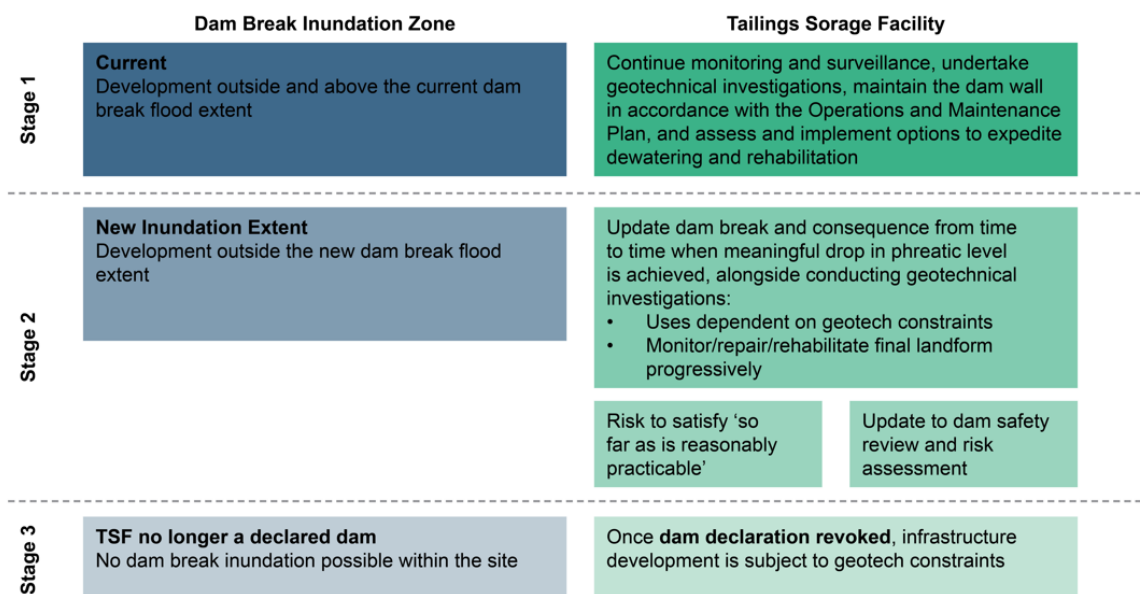
The dam is currently compliant with the recommendations stipulated in various Australian National Committee on Large Dams (ANCOLD) guidelines and Dam Safety NSW guidelines based on the surveillance report from 2025. Dambreak and consequence assessments were undertaken in 2023 and updated in 2024. A safety review of the dam has been undertaken in 2025 followed by risk assessment and 'so far as is reasonably practicable' (SFAIRP) assessment. A staged capping programme is implemented at the dam and is still ongoing.

Until the dam is de-declared, the dam safety monitoring and surveillance activities shall continue to be implemented following ANCOLD guidelines and Dam Safety NSW guidelines. The dam (NSD and TSF) shall continue to be maintained following the dam safety management system (DSMS). Investigations at the TSF shall be undertaken to understand the reduction in phreatic level as well as exact ground water level within the tailings.

New and existing infrastructure, such as roads that traverse the dambreak inundation zone and have the potential to increase societal risk, must be appropriately designed to ensure they are located above the modelled dambreak inundation level whilst the TSF is a declared dam. The inundation extent will reduce over time, with the reduction in phreatic level due to natural infiltration in the TSF, releasing more land progressively available for development subject to geotechnical constraints. Areas above the TSF could provide land uses which do not compromise the capping once rehabilitation is complete and a more stable landform is achieved over time.

Options for accelerated dewatering the TSF can be explored to reduce the timeframe for revocation of declared status.

A flow chart has been prepared and presented in Figure 5-6 with recommendations to support staged development. A list of activities required to be undertaken at the NSD and TSF to continue maintain the safety of the dam discussed in this report is compiled and presented in Table 5-1.



N.B. No development on dam wall and within 10m of dam footprint until de-declared

Figure 5-6 Recommendations to support staged development of the TSF

The overarching aims of the recommendations to be translated into the Master Plan are as follows:

- To enable safe, adaptive, and long-term development of the TSF area by embedding a risk-informed planning framework that recognises and responds to the Tailings Storage Facility and associated dam break hazards over its rehabilitation and post-closure phases until de-declared.
- To maintain a high level of dam safety by continuing regular inspections, monitoring, and maintenance so that any issues are identified early and addressed quickly.
- To regularly update understanding of flood risk from a potential dam failure as conditions improve over time, so that more land can safely be considered for future development.
- To minimise risk to people by carefully controlling development near the TSF, limiting sensitive land uses in affected areas, and ensuring infrastructure is safely designed, until the dam is no longer declared.

Table 5-1 Recommendations

No	Recommendation
Ongoing dam safety management recommendations until TSF is de-declared	
1	Continue undertaking dam safety monitoring and surveillance as specified in the Dam Safety Management System, including: - Routine visual inspections at the dam - Periodic data collection from the monitoring instruments (V-notch, piezometers, deformation targets) - Analysis of collected data - Maintenance activities as required at the dam including the monitoring instruments - Intermediate (annual) and 5-yearly risk review and comprehensive surveillance inspections.
2	Undertake dam break and consequence assessments when a meaningful reduction in the phreatic level in the TSF is achieved. Where an updated assessment has been completed and approved by the relevant authority and demonstrates a reduction in the Dambreak Inundation Zone, new development may be considered permissible on land located outside the revised inundation zone.
3	Undertake periodic review/update of Operation and Maintenance Plan (OMP) and Dam Safety Emergency Plan (DSEP) following Dam Safety NSW requirements.

No	Recommendation
4	Undertake an update to dam safety review and risk assessment as well as the so far as is reasonably practicable (SFAIRP) framework in the next 10 – 15 years. Required site-based investigations shall also be undertaken to support the dam safety review.
Rezoning recommendations whilst the TSF remains a declared dam	
5	Adopt a staged, risk-based land release framework whereby development is only permitted outside the mapped dambreak inundation zone, unless updated technical assessments demonstrate a reduced risk profile.
6	Align future development approval to updated dam break and consequence assessments. Where revised modelling demonstrates a reduction in the inundation extent, additional land may be progressively released for development outside the updated inundation area.
7	Permit only low-intensity and compatible interim uses within the inundation zone (e.g. open space, passive recreation, conservation), ensuring these uses do not obstruct flood pathways or increase hazard levels.
8	Ensure all new and upgraded infrastructure within or crossing the inundation zone: – is located above the modelled dam break inundation level, and – is designed to withstand flood and debris loading without increasing risk or altering flow paths.
9	Maintain a development exclusion and protection buffer around the Northern Saddle Dam (NSD) (minimum 10 m), to preserve access for inspection, maintenance, and to protect embankment integrity. No structures, earthworks, or deep-rooted vegetation are to be permitted within this zone.
10	Restrict development on the TSF landform to uses that: – do not compromise the capping system or drainage – can tolerate ongoing settle and uneven ground conditions – do not introduce water into the tailings material.
11	Require updated technical investigations (including groundwater and phreatic level monitoring) as a mechanism to inform intensification of land uses, confirm improving site conditions and inform refinement of development constraints.
12	Ensure utility infrastructure and services are designed and located to avoid adverse interaction with the TSF, including preventing leakage, infiltration, or loading impacts that could affect stability.
Rezoning recommendations following TSF de-declaration	
13	Require site-specific engineering assessment for any post-declaration development on the TSF, demonstrating that proposed land uses and structures will not adversely affect long-term geotechnical stability.
14	Control intrusive works (e.g. piling, excavation, boreholes) through a requirement for detailed geotechnical assessment and reinstatement measures, ensuring the integrity of the landform is maintained.
15	Ensure final land uses are compatible with residual geotechnical constraints, including settlement potential, ground variability, and long-term stability limitations.
16	Maintain a risk-aware planning approach post-declaration, recognising that while regulatory dam status may be removed, legacy risks associated with the TSF materials and landform persist and must be managed through planning controls and development conditions.

6 References

1. Dam break failure impact assessment (ATCW, 2023).
2. Design Record Report (ATCW, 2025)
3. Insights obtained from internal workshops with DPHI, package leads and agencies.
4. Post Mining Land Use – Deliverability of Pilot Projects – Due Diligence Report – Macquarie Coal Complex, prepared by Aurecon in February 2026.
5. Surveillance review reports (ATCW 2021, 2022, 2023, 2024, 2025).
6. TSF risk assessment and SFAIRP framework report (ATCW, 2025)
7. West Wallsend Dambreak (Presentation Slides) (2024)
8. West Wallsend Tailings Dam Landform Finalisation (WSP, 2025)



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